

RESEARCH DRAFT

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1. HALPHEN SURFACES

Remark 1.1. It is well-known (e.g. as in [DZ99] and [CDL24, Ch.5 §6]) that a Halphen surface of index n is a relative minimal elliptic surface S with F_1 the divisor class of a fibre satisfying $F_1 = -nK_S$. By [DZ99, §2.4], an index 1 Halphen surface admits a section and has no multiple fibres, and an index 2 Halphen surface has a unique double fibre $2F_1$. By [DK24, Prop. 9.1.4], any terminal Coble surface of K3 type is isomorphic to one of the following:

- A blowup of a Halphen surface of index 1 at two singular points of two simple reduced fibres of types $\tilde{A}_k, \tilde{A}_m$ where $k + m \leq 8$, or
- A blowup of a Halphen surface of index 2 at one singular point of a single simple reduced fibre of type $\tilde{A}_n$ where $n \geq 8$

This is used in [DK24, Prop. 9.1.5] to show that $K_S^2 \geq -10$, and since $K_S^2 = -n$, this shows that such Coble surfaces have at most ten boundary components.

Of most immediate relevance to us is the case considered in [Cob19] where $n = 1$. Following [DK24, Ex. 9.1.7], the image $W := V(f_6) := \eta(C)$ is an irreducible plane sextic with 10 A_1 singularities¹. One can find a cubic $V(f_3)$ passing through nine of them and form a pencil $V(\lambda f_6 + \mu f_3^2)$ such that the minimal resolution of its basepoints is a Halphen surface of index 2. The proof of [DK24, Prop. 9.1.8] indicates that the moduli space of general Halphen surfaces of index 2 has dimension 9.

The following is [CDL24, Prop. 3.1]:

Lemma 1.2. *Let X be a Coble surface and $\pi_E : X \rightarrow Y$ be the blowing down of a (-1) -curve E . Then*

- Y is a Halphen surface of index 2;
- C is the proper transform of the fiber F containing $y_0 = \pi_E(E)$;
- the fiber F is irreducible, and y_0 is its unique singular point.

Conversely, the blow-up of a singular point of an irreducible non-multiple fiber of a Halphen surface of index 2 is a Coble surface.

Remark 1.3. Why introduce Halphen surfaces? I conjecture that the geometric construction(s) above yield some kind of correspondence between moduli spaces of Coble surfaces with n boundary components and the various moduli spaces of index 2 Halphen pencils on the $g = 0$ line of Nikulin's triangle diagram of 2-elementary lattices. Regarding the coarse spaces as period domains attached to lattices, the lattices match up precisely; see ???. Applications of mirror moves indicate that the cusp diagrams would correspondingly coincide as well.

¹ A_1 or A_2 singularities.

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